

ZHEJIANG
UNIVERSITY PRESSThe Institution of
Engineering and Technology

WILEY

ORIGINAL RESEARCH PAPER

Fast-Tracker 2.0: Improving autonomy of aerial tracking with active vision and human location regression

Neng Pan^{1,2} | Ruibin Zhang^{1,2} | Tiankai Yang^{1,2} | Can Cui^{1,2} | Chao Xu^{1,2} | Fei Gao^{1,2}

¹State Key Laboratory of Industrial Control Technology, Zhejiang University, Hangzhou, China

²Huzhou Institute of Zhejiang University, Huzhou, China

Correspondence

Can Cui, Huzhou Institute of Zhejiang University, Huzhou 313000, China.
Email: cuiican1990@zju.edu.cn

Funding information

National Natural Science Foundation of China, Grant/Award Numbers: 62003299, 62088101

Abstract

In recent years, several progressive studies promote the development of aerial tracking. One of the representative studies is our previous work Fast-Tracker which is applicable to various challenging tracking scenarios. However, it suffers from two main drawbacks: (1) the oversimplification in target detection by using artificial markers and (2) the contradiction between simultaneous target and environment perception with limited onboard vision. In this study, we upgrade the target detection in Fast-Tracker to detect and localise a human target based on deep learning and non-linear regression to solve the former problem. For the latter one, we equip the quadrotor system with 360° active vision on a customised gimbal camera. Furthermore, we improve the tracking trajectory planning in Fast-Tracker by incorporating an occlusion-aware mechanism that generates observable tracking trajectories. Comprehensive real-world tests confirm the proposed system's robustness and real-time capability. Benchmark comparisons with Fast-Tracker validate that the proposed system presents better tracking performance even when performing more difficult tracking tasks.

The cover image is based on the Original Article *Fast-Tracker 2.0: Improving autonomy of aerial tracking with active vision and human location regression* by Can Cui et al., <https://doi.org/10.1049/csy2.12033>.

KEYWORDS

autonomous aerial vehicles, motion planning, navigation, path planning

1 | INTRODUCTION

Tracking a moving target with flying robots blooms in both industry and academia in recent years. Thanks to the advancement of vision and navigation technology from researches like [1–4], aerial tracking has shown significant robustness and impressive agility. In our previous work, we present a complete aerial tracking system Fast-Tracker [5], which enables a quadrotor to autonomously track an agile target with uncertain intention in unknown cluttered environments and has been successfully applied in versatile scenarios.

However, Fast-Tracker still has severe drawbacks. Chiefly, the artificial marker-based target detection module hinders the

wider application of our system. In this study, we upgrade the target perception as a markerless one, which detects a human target based on deep learning and estimates its location with pre-trained non-linear regression. Besides, as pointed out in Fast-Tracker, it is contradictory to simultaneously detect the target and surrounding obstacles with the limited field of view (FoV). To resolve this, we replace the camera array mechanism in Fast-Tracker with a lightweight gimbal onboard the quadrotor (as shown in Figure 1) to separate the vision for tracking and for environment sensing. The conflict between the target and environment has been resolved with the actively controllable gimbal camera, and the quadrotor itself focusses more on safe navigation. In addition, we incorporate an occlusion

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

© 2021 The Authors. *IET Cyber-Systems and Robotics* published by John Wiley & Sons Ltd on behalf of Zhejiang University Press.

penalty in our path-searching method that encourages the quadrotor to find observable paths while tracking.

On the whole, our systematic solution consists of five steps in a cascaded way: (1) the detection and location estimation of a human target (Section 5); (2) the prediction of target future motion without its motion intention (Section 5.1); (3) the tracking path searching (Section 5.2) with an occlusion-aware mechanism; (4) the generation of safe and dynamically feasible tracking trajectories (Section 5.3); and (5) the construction of an onboard system (Section 6) with omnidirectional vision. The proposed solution addresses most of the major challenges in aerial tracking.

The proposed methods are integrated into a quadrotor system (as shown in Figure 2). Sufficient experiments in difficult real-world environments validate that the system can accurately detect and localise a human target and then

aggressively and safely track it. We also compare this work with Fast-Tracker. The results show that even the target detection and localisation task become more challenging. The proposed system still achieves better tracking performance. Contributions of this study includes:

- A lightweight and intention-free human target detection and localisation method based on deep learning and non-linear regression.
- Improvement of the integrated online tracking planning module in Fast-Tracker by incorporating an occlusion-aware mechanism.
- The machine design of the customised free-rotating gimbal camera to widen the quadrotor's field of view (FoV).
- An onboard quadrotor system that integrates the proposed methods and the presentation of the systematic solution with extensive tests and evaluations.

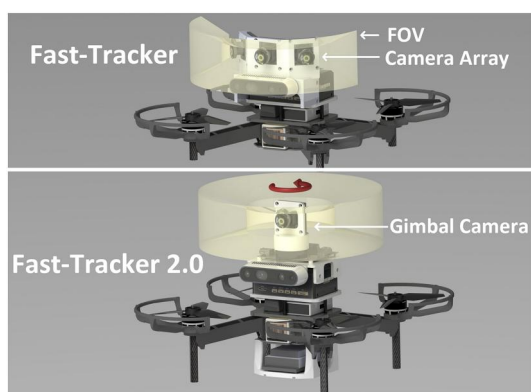


FIGURE 1 Comparison of the hardware design between the proposed and the previous quadrotor systems. Compared with the fixed field of view (FoV) in Fast-Tracker, the proposed system enables free rotation of the camera

2 | RELATED WORK

2.1 | Aerial tracking systems

Some previous studies [6–8] develop aerial tracking systems based on vision-based controllers that take the distance error from the quadrotor to the target obtained from the image space as feedback. These methods fail to consider safety constraints, thus limiting the tracking scenarios only in wide-open areas. In recent years, several studies present more complete aerial tracking systems that consider obstacle avoidance, dynamical feasibility, and tracking efficiency. Nægeli et al. [1] rely on ground truth of the target's location and set a strong assumption on the obstacle's shape, limiting the application scenarios to artificial environments. Bonatti et al. [2] use a

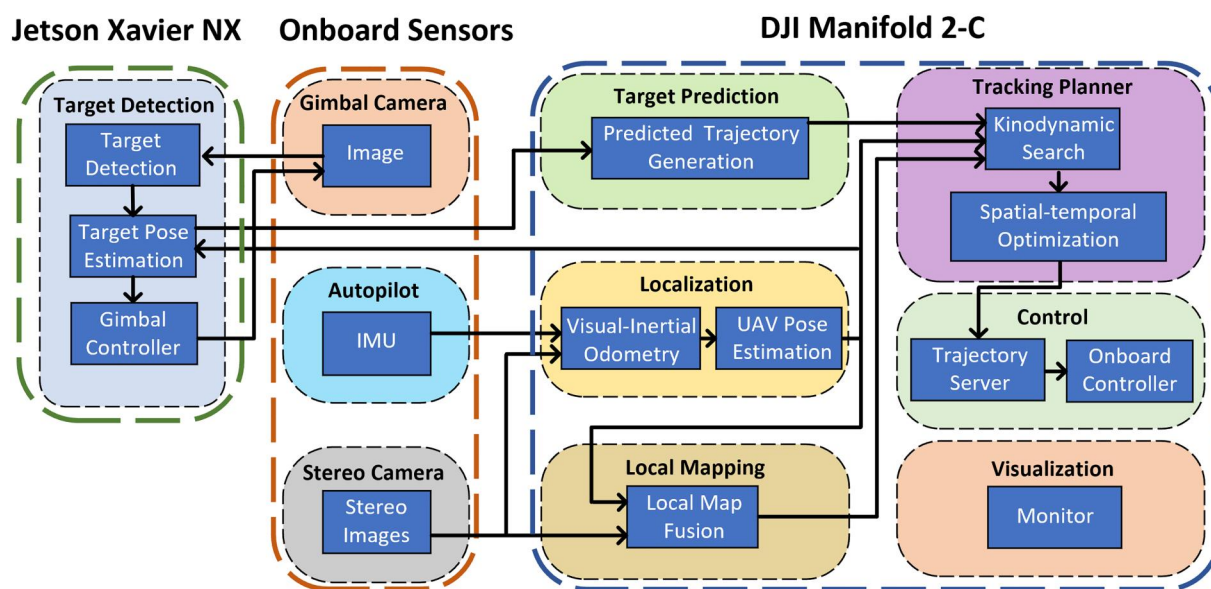


FIGURE 2 Software architecture of our quadrotor system. The sensing, planning and controlling modules run parallelly using onboard computing resources without external sensors

long-range LiDAR sensor for mapping and use GPS for localisation. However, bulky LiDAR sensors and GPS-based sensors can only be deployed on large outdoor platforms. Such system settings cannot work properly in indoor or cluttered environments like Fast-Tracker does. Jeon et al. [3] require a prebuilt environmental map and assume that the target moves through a set of given via-points, making the system hard to be applied out of laboratories. Chen et al. [4] and Fast-Tracker are able to handle unknown obstacle information and free-moving targets. Nevertheless, the two systems depend on handcrafted marker-based target detection, which oversimplifies the detection task. Wang et al. [9] took the occlusion of the target into consideration and conducted impressive experiments. However, such methods still fail when the target moves extremely agilely due to the limited FoV. In short, the above aerial tracking systems are not applicable to challenging field applications.

2.2 | Target prediction and tracking trajectory planning

The prediction module estimates the target's future motion as a guidance for trajectory planning. Similar schemes are proposed in [1, 2, 7] that adopt Kalman Filter with a specific motion model, which is not accurate enough to precisely predict the future motion of a target with uncertain intention. Jeon et al. [3] utilise covariant optimisation that incorporates collision avoidance for target motion prediction. However, the collision term in the objective is computed from the Euclidean signed distance field (ESDF), consuming substantial computing resources. To estimate the target's future motion, Chen et al. [4] adopt polynomial regression, which is formulated as quadratic programming (QP). Fast-Tracker improves this method by adding dynamic constraints and time-variant confidence in the target-predicted trajectory optimisation.

As for tracking trajectory planning, Nægeli et al. [1] propose a receding horizon Model Predictive Control (MPC) planner that considers visibility to targets, collision avoidance, and occlusion minimisation. Similarly, the planning module in [2] optimises a set of cost functions based on covariant gradient descent. However, the optimisation formulations both contain multiple non-convex terms, and the solution might fall into a local minimum. Jeon et al. [3] propose a bi-level approach composed of a graph-search based path planner and a hard-constrained trajectory optimiser. Because a dense local graph has to be built and traversed in every search process, the graph-search method is highly time-consuming. Chen et al. [4] optimise the tracking error and high-order derivatives of the trajectory within a flight corridor. Nevertheless, this work fails to account for occlusion against obstacles.

2.3 | Human detection and localisation

Several previous studies [2, 8, 10] focus on detecting two-dimensional (2D) targets utilising monocular images and

recovering their three-dimensional (3D) location. Bonatti et al. [2] combine MobileNet [11] and Faster-Region-based Convolutional Neural Network (RCNN) [12] for target detection. A ray-casting module is then proposed to project the 2D bounding box onto a height map provided by an onboard LiDAR sensor. As stated above, the LiDAR sensor is a heavy load in target-tracking scenarios. Cheng et al. [8] set a tracking scenario that the quadrotor flies at a high altitude and subsequently considers the human target as a point. The quadrotor's altitude is then approximated as the target's depth in the images and utilised for 3D location estimation. However, the high-altitude flight prevents the method from applying to indoor or narrow environments. Huang et al. [10] use a sequence-to-sequence network to estimate 3D pose from a sequence of 2D joints obtained from skeleton detection and then define an optimisation function to estimate the target's location. The above methods are mainly used in aerial cinematography, where target pose estimation is required for planning the shooting angle. As for aerial tracking, there is no pressing need for target pose estimation. Compared with [10], although we use the skeleton detection method as well, we directly estimate the location of the target without pose estimation to save computing resources.

To enable active vision and improve noisy target observations, previous studies [1, 2, 8, 10, 13, 14] equip an onboard 3-degree of freedom (DOF) gimbal camera. The target can be kept at the desired position within the camera plane by adopting closed-loop controllers or optimisation methods. In fact, due to the fact that the target does not move drastically in the vertical direction in most tracking scenarios, controlling its horizontal position in the camera plane is sufficient to avoid target loss. We thus customise a lightweight 1-DOF gimbal instead, ensuring the agility of the proposed quadrotor system. We also equip an electrical slip ring structure to enable free rotation of the gimbal to enhance target perception capability.

3 | OVERVIEW

3.1 | Problem formulation

The overall task is to control a quadrotor to detect, localise, and track a human target with uncertain intention in unknown environments. Let $\hat{\mathbf{p}}_t^w = \{\mathbf{x}_t^w, \mathbf{y}_t^w, \mathbf{z}_t^w\} \in \mathbb{R}^3$ be the estimated 3D position of the target in the world frame (Section 4), and $\hat{P}_t(t) : (t_c, t_p] \rightarrow \mathbb{R}^3$ be the target-predicted trajectory from the current time t_c to t_p (Section 5.1). For the quadrotor planning module, we use voxel occupancy grids denoted as $\mathcal{G} : \mathbb{R}^3 \rightarrow [0, 1]$ to represent the environment. Using $\hat{P}_t$ as a guidance, the front-end path-searching method generates a grid path with kinodynamic information (Section 5.2). Utilising the end-derivatives and the flight corridor $\mathcal{F}$ obtained from the grid path, the back-end optimiser generates a piecewise-polynomial trajectory $P_q(t) \in \mathbb{R}^3$ for the quadrotor to track the target (Section 5.3).

For the tracking scenarios, we make a set of simplifying assumptions:

- The target's motion in the vertical direction is negligible during the tracking process.
- The upper body length of the target remains roughly the same.
- The target moves smoothly with bounded velocity and acceleration.

3.2 | Active vision design

According to assumption a, we focus on controlling the target's horizontal position in the camera plane. A 1-DOF gimbal is thus deployed to control the camera's yaw angle so that the target is positioned in the middle of the frame in the horizontal direction. We use a PI controller to control the camera independently of the quadrotor's motion planning.

Since the camera and the gimbal are fixed, the camera's connection cable will limit its rotation. To enable the gimbal motor's free rotation, we replace the connection cable with a customised electrical slip ring. A slip ring's primary components are a rotating ring and stationary brushes that rub with the ring, as illustrated in Figure 3. Data transmission between the spinning and stationary cables is realised for they are connected with the rotating ring and the brushes, respectively. Therefore, the camera can rotate freely while remaining connected with the onboard computer.

Credit to the active vision, we now keep the quadrotor facing the direction of its linear velocity rather than facing the target. As a result, the target and the surrounding obstacles can be well observed at the same time. As a result, the quadrotor is able to keep trace of a moving target in complex environment as shown in Figure 4.

4 | HUMAN DETECTION WITH LOCATION REGRESSION

This section elaborates on the human target detection and localisation methods. Utilising monocular images obtained from the gimbal camera, this module outputs $\hat{\mathbf{p}}_t^w = \{x_t^w, y_t^w, z_t^w\}$. The entire module is shown in Figure 5.

4.1 | Human target detection

For human detection, we apply OpenPose [15], a deep learning-based 2D pose estimation method, which detects 2D joints, namely the head, shoulders, knees etc. OpenPose adopts a bottom-up parsing architecture that jointly learn part locations and their association via part affinity fields (PAFs) and presents remarkable performance and efficiency. We equip OpenPose with TensorRT [16] inference optimiser to ensure real-time capability on the onboard platform.

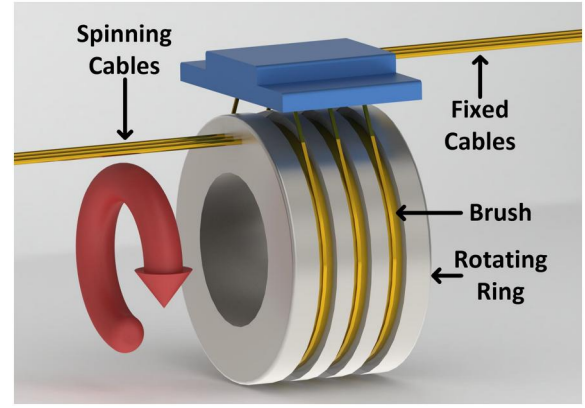


FIGURE 3 The schematic diagram of the slip ring



FIGURE 4 The outdoor experiment in a dense forest

Due to assumption b, the target's upper body length can be utilised as an invariant feature. We take the distance from the hip to the neck in the camera plane as the upper body length and utilise it for target localisation.

4.2 | Non-linear regression-based target localisation

Let L_t be the detected upper body length in the camera plane. We also utilise the u coordinate value of the target's centre in the camera plane denoted as u_t for localisation. Non-linear functions are used to establish the mapping from L_t and u_t to the estimated target location in the camera frame denoted as $\hat{\mathbf{p}}_t^c = \{x_t^c, y_t^c, z_t^c\}$. Especially, z_t^c is set to be a constant because of assumption a. We then define the following functions with a set of undetermined parameters:

$$\begin{cases} x_t^c = \lambda_1 e^{k_1 L_t} + \lambda_2 e^{k_2 L_t}, \\ y_t^c = \left(\lambda_3 e^{k_3 u_t} + \lambda_4 e^{k_4 u_t} \right) (a x_t^c + b). \end{cases} \quad (1)$$

The adjusting of the undetermined parameters can be done by a non-linear fitting process and common software like MATLAB is qualified for this.

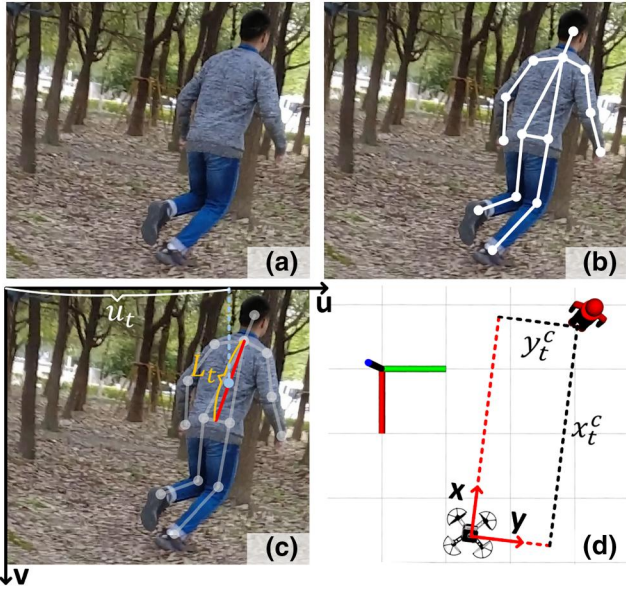


FIGURE 5 Example of human detection in a realistic scenario. (a) A raw monocular image for target detection. (b) The target detection result. (c) The extracted features L_t and u_t in the camera plane. (d) The 3D location estimation result

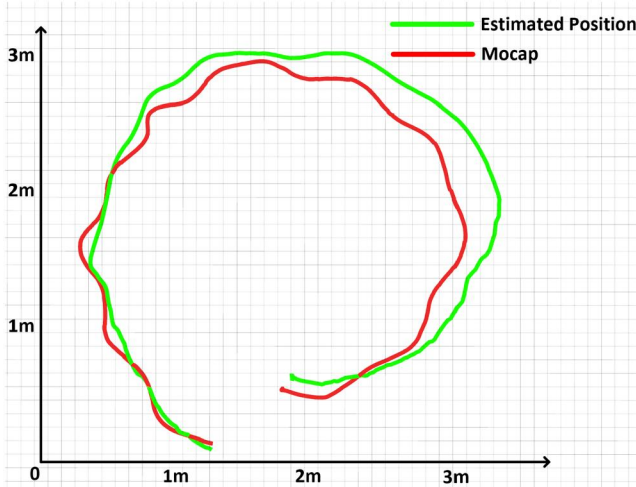


FIGURE 6 Tests of the target localisation method under the mocap system

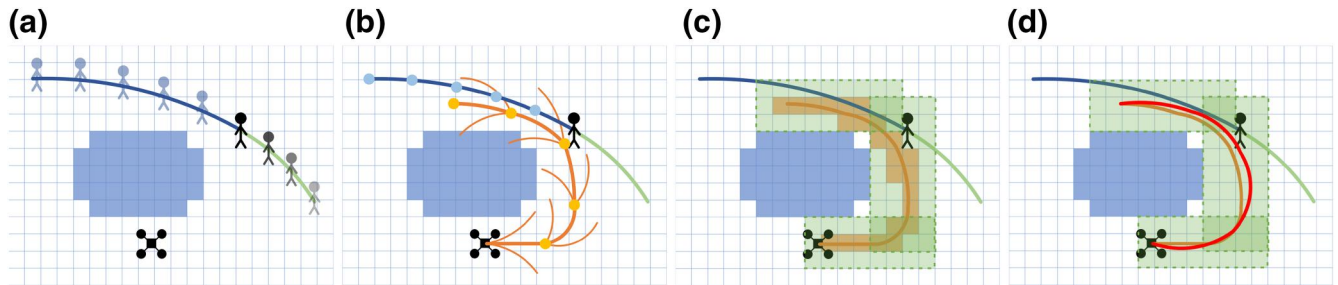


FIGURE 7 Procedures of the tracking planning module. (a) A target-predicted trajectory $\hat{P}_t(t)$ (the blue curve) is generated based on past location estimations $\hat{p}_t^w$ (the black markers). (b) A kinodynamic grid path consisted of motion primitives (the orange curves) is generated for the quadrotor to track. (c) A flight corridor $\mathcal{F}$ (the sequence of green cubes) is generated based on the grid path. (d) A spatial-temporal trajectory P_q (the red curve) is generated within the flight corridor

In practice, we make a dataset under the ground truth obtained from a motion capture system (mocap) to train these parameters. $\hat{p}_t^c$ is converted to $\hat{p}_t^w$ through coordinate transformation in the end.

We test the performance of the location regression method, as shown in Figure 6. The average distance error is about 20 cm, which is acceptable for tracking scenarios. The results (Section 6) also confirm that this method is practical and effective.

5 | ONLINE TRACKING PLANNING

In this section, we introduce the tracking planning module of Fast-Tracker shown in Figure 7 and the proposed improvements on it (Figure 8).

5.1 | Target motion prediction

This method updates the target predicted trajectory $\hat{P}_t(t) : (t_c, t_p]$ utilising the past target estimations acquired from Section 4. Polynomial regression is adopted to fit the target motion estimations. We specially use Bernstein basis polynomial (Bézier curve) to enforce dynamical constraints due to its convex hull property and hodograph property (described in [17]). The target-predicted trajectory is presented as:

$$\hat{P}_t(t) = \sum_{i=0}^n c_i b_n^i(t) \quad (2)$$

where each $b_n^i(t) = \binom{n}{i} \cdot t^i \cdot (1-t)^{n-i}$ is an n -degree Bernstein polynomial basis, and $[c_0, c_1, \dots, c_n]$ is the set of control points of the Bézier curve.

The object function J_{pre} is formulated as QP that includes the residual term J_d and regularisation term J_r :

$$J_{\text{pre}} = J_d + J_r, \quad (3)$$

where J_d minimises the distance residual between $\hat{P}_t$ and the past target location estimations. J_r is an acceleration regulator to avoid overfitting.

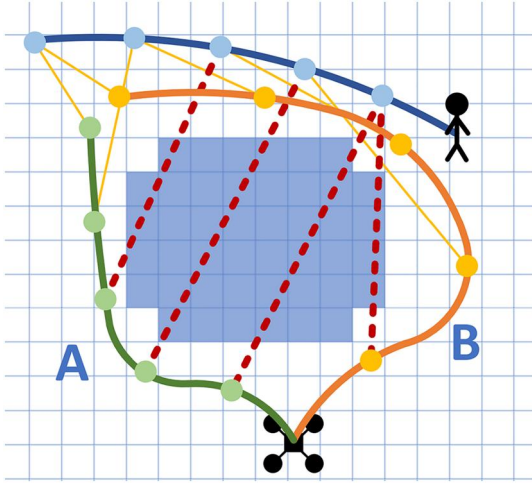


FIGURE 8 Illustration of the occlusion-aware mechanism. Two topology paths (the green and orange curves) are generated based on the target-predicted trajectory (the blue curve). Although topology A is shorter than B, it is suboptimal because it will cause more occlusion than B in the future tracking process. Credit to the occlusion penalty, topology B will be chosen

Based on assumption c, the predicted velocity and acceleration are constrained in pre-defined ranges to ensure the dynamical feasibility. Time-variant weights are also added to decrease the confidence of old target location estimations. The obtained trajectory is then extrapolated to $(t_c, t_p]$ for target motion prediction. The detailed formulations are described in [5].

A target-relocating strategy is proposed in [5] to make the quadrotor actively explore and rediscover the target after losing it. With the active vision, we keep the gimbal rotating to search for the target during the relocating process. This mechanism improves the target-relocating strategy's effectiveness and is presented in the results (Section 6).

5.2 | Occlusion-aware tracking path searching

The path-searching method is originated from the hybrid A* algorithm [18]. Let the position and velocity vector $\mathbf{x} = (p_x, p_y, p_z, v_x, v_y, v_z)^T$ be the state of the quadrotor. The quadrotor's acceleration is taken as the control input $\mathbf{u}$ and discretised to generate motion primitives, namely the nodes in the hybrid A* algorithm.

The cost function of each node is written as $f_c = g_c + h_c$, where g_c represents the actual cost from the initial state X_0 to the current expanded state $\mathbf{x}_c$ and h_c means the heuristic cost. g_c balances the control-effort $\mathbf{u}$ and the time T of a trajectory by minimising an energy-time cost defined as:

$$J_t = \int_0^T \|\mathbf{u}(\tau)\|^2 d\tau + \rho T, \quad (4)$$

where ρ is the weight term. In practice, this cost function is discretised to be the sum of the expanded nodes' cost.

h_c considers the heuristic cost from $\mathbf{x}_c$ to the goal state $\mathbf{x}_g$. We denote the dynamic cost from $\mathbf{x}_c$ to $\mathbf{x}_g$ as $D(\mathbf{x}_c, \mathbf{x}_g)$. It is obtained from solving an optimal boundary value problem (OBVP) proposed in [19]. To make the path search both reliable and forward-looking, we use the weighted value of the target's current state $\mathbf{x}_{tc}$ and predicted state $\mathbf{x}_{tp}$ obtained from $\hat{P}_t$ as the goal state $\mathbf{x}_g$. Moreover, a time penalty P_{time} is added to h_c to speed up searching.

In this study, occlusion check is conducted between $\mathbf{x}_c$ and $\mathbf{x}_{tp}$ in each expanding process. We claim that occlusion will happen if the line from $\mathbf{x}_c$ to $\mathbf{x}_{tp}$ collides with obstacles, and this node cannot pass the occlusion check. An additional term P_{occ} is then added to h_c to penalise such nodes. This mechanism is illustrated in Figure 7. In general, h_c is formulated as follows:

$$h_c = D(\mathbf{x}_c, \mathbf{x}_g) + P_{\text{time}} + P_{\text{occ}}, \quad (5)$$

$$P_{\text{time}} = K_t(S_t - t_c), \quad (6)$$

$$P_{\text{occ}} = \begin{cases} K_{\text{occ}} & \text{if the line between } X_c \text{ and } X_g \text{ collides} \\ 0 & \text{else} \end{cases}, \quad (7)$$

where K_{occ} and K_t are the weight terms, and S_t is the sum of expected expansion time, which can be obtained from the uniform speed model for example.

5.3 | Safe tracking trajectory generation

We use the efficient spatial-temporal piece-wise polynomial trajectory generator [20] for back-end optimisation. This method derives an analytic gradient for the waypoints q_w and piece times T with linear complexity. Leveraging the exact gradient, the decision variables are reduced to q_w and T .

Chiefly, a flight corridor $\mathcal{F}$ is generated based on the grid path obtained from the path-searching method, defined as:

$$\mathcal{F} = \cup_{i=1}^M \mathcal{C}_i, \quad (8)$$

where each $\mathcal{C}_i$ is a finite cube:

$$\mathcal{C}_i = \{x \in \mathbb{R}^3 | \mathbf{A}_c x \leq b_{c_i}\}, \quad (9)$$

For more details about generating the flight corridor, please refer to [17].

For such a $\mathcal{F}$, the start and goal position is located in $\mathcal{C}_1$ and $\mathcal{C}_M$, respectively. This method assigns each intermediate waypoint q_{wi} in the intersection $\mathcal{C}_i \cap \mathcal{C}_{i+1}$ to ensure safety.

The optimisation formulation trades off between the smoothness J_S , the corridor constraint J_F , and the aggressiveness penalty J_D :

$$J_{\Sigma}(q, T) = J_S(q, T) + J_F(q, T) + J_D(q, T). \quad (10)$$

J_S minimises the magnitude of jerk along the piecewise trajectory in 3 dimensions to improve smoothness and is defined as:

$$J_S(q, T) = \sum_{i=1}^M \int_{T_{i-1}}^{T_i} \mathbf{u}_i(t)^T \mathbf{u}_i(t) dt, \quad (11)$$

$$s.t. \mathbf{u}_i(t) = \mathbf{q}_i^{(3)}(t), \forall t \in [T_{i-1}, T_i]. \quad (12)$$

J_F is a logarithmic barrier term to ensure that each $\mathbf{q}_{w_i}$ is constrained in $\mathcal{C}_1 \cap \mathcal{C}_{i+1}$, which is defined as:

$$J_F(q_w) = -\kappa \sum_{i=1}^{M-1} \sum_{j=i}^{i+1} 1^T \ln[b_{C_j} - \mathbf{A}_{C_j} \mathbf{q}_{w_i}]. \quad (13)$$

where κ is a constant coefficient, $\mathbf{1}$ denotes an all-ones vector and $\ln[\cdot]$ denotes the entry-wise natural logarithm.

The last term J_D adjusts the aggressiveness of the whole trajectory. It is defined as:

$$J_D(q_w, T) = \rho_t \sum_{i=1}^M T_i + \rho_v \sum_{i=1}^{M-1} l\left(\left\|\frac{\mathbf{q}_{w_{i+1}} - \mathbf{q}_{w_i}}{T_{i+1} + T_i}\right\|^2 - v_m^2\right) + \rho_a \sum_{i=1}^{M-1} l\left(\left\|\frac{\frac{(\mathbf{q}_{w_{i+1}} - \mathbf{q}_{w_i})}{T_{i+1}} - \frac{(\mathbf{q}_{w_i} - \mathbf{q}_{w_{i-1}})}{T_i}}{\frac{(T_{i+1} + T_i)}{2}}\right\|^2 - a_m^2\right), \quad (14)$$

where $l(x) = \max(x, 0)^3$, v_m is the maximum velocity and a_m is the maximum acceleration. ρ_t prevents the entire duration from growing too large. ρ_v and ρ_a limit the velocity and acceleration, respectively, to prevent the tracking trajectory from being too aggressive.

6 | RESULTS

6.1 | Implementation details

The quadrotor system built for real-world experiments is shown in Figure 9. It is equipped with Intel RealSense D435 depth camera for ego-localisation and mapping. The localisation, mapping, and motion planning algorithms run on Manifold2 with i7-8550U central processing unit (CPU), while the target detection algorithm runs on Jetson Xavier NX with the Graphics processing unit (GPU) performance of 21 TOPS. The horizontal gimbal camera is driven by a brushless motor. The quadrotor is

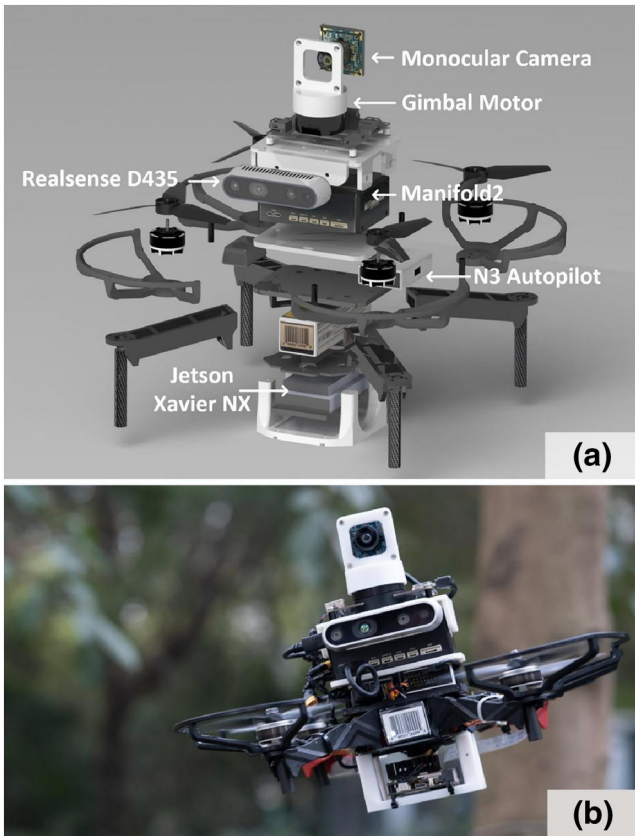


FIGURE 9 Hardware platform. (a) Exploded diagram of the hardware platform. (b) Real-world appearance of the system

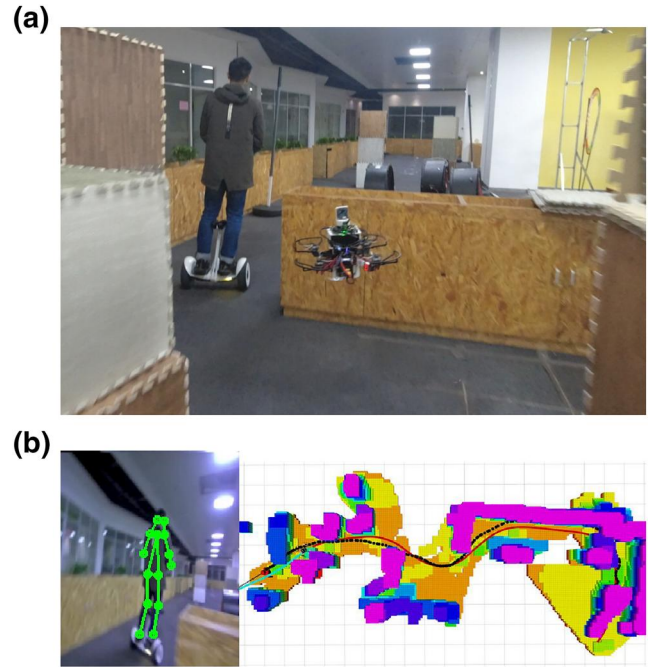


FIGURE 10 Demonstration of the indoor experiment. (a) The indoor experiment in a narrow and cluttered corridor. (b) The third perspective and the visualisation of the indoor experiment

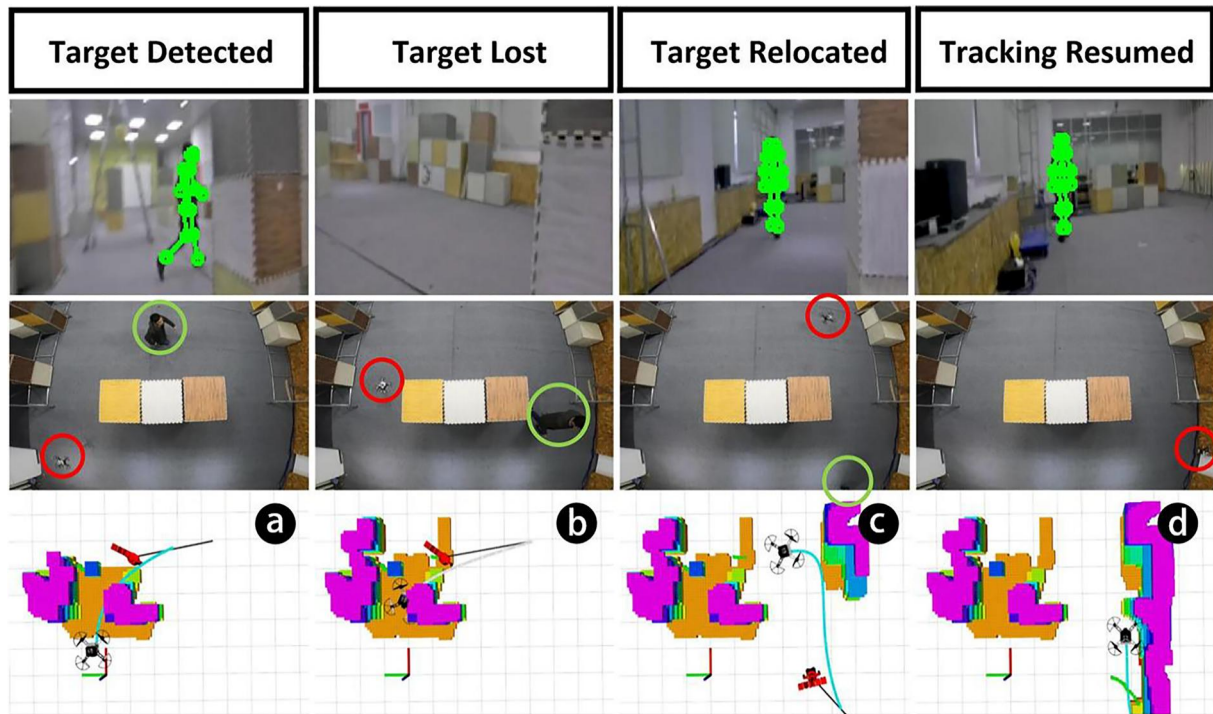


FIGURE 11 The schematic diagram of the target relocation strategy. In the bottom visualisation diagrams, the black curves represent the target-predicted trajectory, and the cyan curves represent the quadrotor's tracking trajectory. (a) The quadrotor tracks the target based on target location estimation and motion prediction. (b) After the obstacles occlude the target, the quadrotor flies along the re-locating trajectory (gray curve). Meanwhile, the gimbal camera keeps rotating to search for the target. (c) The target is re-located by the gimbal camera. (d) A new trajectory is generated for the quadrotor to carry on tracking the target

300 × 300 mm in size and only weighs 1.4 kg including a 4-s battery, which is dexterous enough for most tracking scenarios. The average computing times are 40 and 20 ms, respectively, for the target detection and motion planning modules. The re-planning frequency of the whole pipeline is set to be 13 Hz.

6.2 | Real-world experiments

This section presents extensive experiments in several unknown cluttered environments to test the proposed approach's robustness and tracking performance.

The first experiment is carried out in a narrow and cluttered corridor, as shown in Figure 10. The target drives a balance car to move agilely in the corridor. The cramped environment requires the quadrotor to respond quickly and re-plan the tracking trajectory in case of an emergency. As a result, the quadrotor flies safely in this narrow environment and closely follows the target.

Another experiment demonstrates the improved target-relocating strategy, as shown in Figure 11. In this experiment, the quadrotor loses the target due to its sudden turn. Moreover, the target turns again and moves fast in another direction, making it hard to be rediscovered. As a result, the target is still relocated because the gimbal camera keeps

searching for the target in all directions when the quadrotor flies along the relocating trajectory.

We also conduct an outdoor experiment in a dense forest, as illustrated in Figure 2. The target zooms through massive trees. Strong winds throw up leaves and dust, making the ego-localisation and map less credible. In spite of this, it turns out that the quadrotor still reaches 3 m/s and closely follows the target in such a challenging environment.

From the experiments, we conclude that the proposed system maintains robustness in challenging tasks and present satisfactory tracking performance as well. More details are available in the attached video.

6.3 | Benchmark comparisons

We compare the proposed system with Fast-Tracker. We set a tracking scenario with sharp turnings, as illustrated in Figure 12. When the target suddenly turns, the surrounding obstacles are likely to cause occlusion. 10 tests are conducted when the target moves in low speed (0.85 m/s) and high speed (1.5 m/s), respectively. A test fails when the quadrotor cannot follow the target all the way. Table 1 reveals that even if the proposed system faces more difficult detection tasks than Fast-Tracker, it still surpasses Fast-Tracker in the tracking



FIGURE 12 Visualisation of the experimental scenario in benchmark comparisons. The sharp turning and massive obstacles are likely to cause occlusion, making this scenario challenging

TABLE 1 Comparison in the sharp-turning scenario

Scenario	Method	Success rate
Low speed	Proposed	10/10
	Fast-Tracker	8/10
High speed	Proposed	9/10
	Fast-Tracker	3/10

performance. With active vision, the target is kept to be detected even when performing jerky motions. Furthermore, the occlusion-aware path-searching method tends to search for grid paths with better visibility.

7 | CONCLUSION

In this study, we improve our previous work Fast-Tracker to apply it in human target-tracking scenarios. We propose a lightweight and practical target detection and localisation method utilising monocular images. To avoid occlusion against obstacles, we implement active vision on a customised gimbal camera, together with an occlusion-aware mechanism in the motion planning module. Extensive real-world experiments confirm that the proposed approach can demonstrate real-time performance on an onboard platform and maintain robustness in complex environments. Furthermore, comparisons with Fast-Tracker reveal that the proposed active vision and occlusion-aware mechanism reinforce target perception capability and make the proposed system applicable to more challenging tracking scenarios.

In the future, we will focus on artistic decision-making in aerial tracking, such as the autonomous selection of shooting angles and relative motion patterns under different scene contexts. Moreover, we will try to apply the proposed approach to other types of unmanned aerial vehicles so as to handle more challenging tracking scenarios.

ACKNOWLEDGEMENTS

This work was supported by the National Natural Science Foundation of China under Grant Nos. 62003299 and 62088101.

CONFLICT OF INTEREST

There is no conflict of interest.

DATA AVAILABILITY STATEMENT

All data included in this study are available upon request by contact with the corresponding author.

ORCID

Neng Pan  <https://orcid.org/0000-0002-2905-2467>

REFERENCES

- Nägeli, T., et al.: Real-time motion planning for aerial videography with dynamic obstacle avoidance and viewpoint optimization. *IEEE Robot. Autom. Lett.* 2(3), 1696–1703 (2017)
- Bonatti, R., et al.: Autonomous aerial cinematography in unstructured environments with learned artistic decision-making. *J. Field Robot.* 37(4), 606–641 (2020)
- Jeon, B., Lee, Y., Kim, H.J.: Integrated motion planner for real-time aerial videography with a drone in a dense environment. In: 2020 IEEE International Conference on Robotics and Automation (ICRA), pp. 1243–1249 (2020)
- Chen, J., Liu, T., Shen, S.: Tracking a moving target in cluttered environments using a quadrotor. In: 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 446–453 (2016)
- Han, Z., et al.: A robust aerial system for tracking agile target in cluttered environments. *arXiv preprint arXiv:2011.03968* (2020)
- Kim, J., Shim, D.H.: A vision-based target tracking control system of a quadrotor by using a tablet computer. In: 2013 International Conference on Unmanned Aircraft Systems (ICUAS), pp. 1165–1172 (2013)
- Kendall, A.G., Salvapantula, N.N., Stol, K.A.: On-board object tracking control of a quadcopter with monocular vision. In: 2014 International Conference on Unmanned Aircraft Systems (ICUAS), pp. 404–411 (2014)
- Cheng, H., et al.: An autonomous vision-based target tracking system for rotorcraft unmanned aerial vehicles. In: 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 1732–1738 (2017)
- Huang, C., et al.: Through-the-lens drone filming. In: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 4692–4699 (2018)
- Howard, A.G., et al.: Mobilenets: efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861* (2017)
- Ren, S., et al.: Faster R-CNN: towards real-time object detection with region proposal networks. *Neural Inf. Process. Syst.* (2015). https://proceedings.neurips.cc/paper/2015/file/14bfa6bb14875e45bba028a21e_d38046-Paper.pdf
- Galvane, Q., et al.: Directing cinematographic drones. *ACM Trans. Graph.* 37(3) (2018)
- Joubert, N., et al.: Towards a drone cinematographer: guiding quadrotor cameras using visual composition principles. *arXiv preprint arXiv:1712.04216* (2016)
- Cao, Z., et al.: Openpose: realtime multi-person 2D pose estimation using part affinity fields. In: IEEE Transactions on Pattern Analysis and Machine Intelligence (2019)
- Gao, F., et al.: Online safe trajectory generation for quadrotors using fast marching method and Bernstein basis polynomial. In: 2018 IEEE

- International Conference on Robotics and Automation (ICRA), pp. 344–351 (2018)
16. TensorRT open source software. <https://github.com/NVIDIA/TensorRT>, 2021
 17. Dolgov, D., et al.: Practical search techniques in path planning for autonomous driving. In: International Symposium on Combinatorial Search, SoCS 2008 (2008)
 18. Zhou, B., et al.: Robust and efficient quadrotor trajectory generation for fast autonomous flight. *IEEE Robot. Autom. Lett.* 4(4), 3529–3536 (2019)
 19. Wang, Z., et al.: Generating large-scale trajectories efficiently using double descriptions of polynomials. *arXiv preprint arXiv:2011.02662* (2020)
 20. Qin, T., Li, P., Shen, S.: Vins-mono: a robust and versatile monocular visual-inertial state estimator. *IEEE Trans. Robot.* 34(4), 1004–1020 (2018)

How to cite this article: Pan, N., et al.: Fast-Tracker 2.0: Improving autonomy of aerial tracking with active vision and human location regression. *IET Cyber-Syst. Robot.* 3(4), 292–301 (2021). <https://doi.org/10.1049/csy2.12033>